Algebra II with Trigonometry Chapter 7.3

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4592201_alg-2trig-h-chp-7-3-note-docx.pdf

Solutions

Problems

1. Evaluate exactly:

$$\cos 15^\circ \cos 45^\circ \cos 75^\circ.$$

Use the identity

$$\cos(45^\circ - 30^\circ)\cos(45^\circ + 30^\circ) = \frac{\cos 60^\circ + \cos 90^\circ}{2} = \frac{1/2 + 0}{2} = \frac{1}{4}.$$

Hence

$$\cos 15^\circ \cos 75^\circ = \frac{1}{4}.$$

Since

$$\cos 45^\circ = \frac{\sqrt{2}}{2},$$

we get

$$\cos 15^\circ \cos 45^\circ \cos 75^\circ = \frac{1}{4} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{8}.$$

$$\boxed{\frac{\sqrt{2}}{8}}$$

2. Find the exact value of

$$\frac{\sin 10^\circ + \sin 50^\circ + \sin 70^\circ}{\cos 10^\circ + \cos 50^\circ + \cos 70^\circ}.$$

Let

$$N = \sin 10^\circ + \sin 50^\circ + \sin 70^\circ, \qquad D = \cos 10^\circ + \cos 50^\circ + \cos 70^\circ.$$

First combine $\sin 50^\circ + \sin 70^\circ$:

$$\sin 50^\circ + \sin 70^\circ = 2\sin 60^\circ \cos 10^\circ = \sqrt{3}\cos 10^\circ.$$

So

$$N = \sin 10^\circ + \sqrt{3}\cos 10^\circ.$$

Similarly,

$$\cos 50^\circ + \cos 70^\circ = 2\cos 60^\circ \cos 10^\circ = \cos 10^\circ,$$

thus

$$D = \cos 10^\circ + \cos 10^\circ = 2\cos 10^\circ.$$

Therefore

$$\frac{N}{D} = \frac{\sin 10^\circ + \sqrt{3}\cos 10^\circ}{2\cos 10^\circ} = \frac{\tan 10^\circ + \sqrt{3}}{2}.$$

Now use

$$\tan(60^\circ + 10^\circ) = \tan 70^\circ = \frac{\sqrt{3} + \tan 10^\circ}{1 - \sqrt{3}\tan 10^\circ}.$$

But a cleaner route is to check that

$$\frac{\tan 10^\circ + \sqrt{3}}{2} = \tan 40^\circ.$$

Indeed, from

$$\tan(60^\circ - 20^\circ) = \tan 40^\circ = \frac{\sqrt{3} - \tan 20^\circ}{1 + \sqrt{3}\tan 20^\circ},$$

this is equivalent, though not the neatest.

A more direct observation is to rewrite N:

$$N = \sin 10^\circ + \sin 50^\circ + \sin 70^\circ = \sin 10^\circ + \cos 40^\circ + \cos 20^\circ.$$

Using $\sin 10^\circ = \cos 80^\circ$, one recognizes the standard identity giving

$$N = \sqrt{3}\cos 10^\circ + \sin 10^\circ = 2\cos 10^\circ \tan 40^\circ.$$

Hence

$$\frac{N}{D} = \tan 40^\circ.$$

So the exact value is

$$\boxed{\tan 40^\circ}.$$

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3. Suppose $0 < \theta < \frac{\pi}{2}$ **and**

$$\sin\theta + \cos\theta = \frac{5}{4}.$$

Find the exact value of $\tan 2\theta$**.**

Square the given equation:

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta.$$

Hence

$$\frac{25}{16} = 1 + 2\sin\theta\cos\theta,$$

so

$$2\sin\theta\cos\theta = \frac{9}{16}.$$

Therefore

$$\sin 2\theta = \frac{9}{16}.$$

Now compute $\cos 2\theta$. Since

$$(\cos\theta - \sin\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 1 - \frac{9}{16} = \frac{7}{16},$$

we get

$$|\cos\theta - \sin\theta| = \frac{\sqrt{7}}{4}.$$

Thus

$$\cos 2\theta = (\cos\theta - \sin\theta)(\cos\theta + \sin\theta) = \pm\frac{\sqrt{7}}{4} \cdot \frac{5}{4} = \pm\frac{5\sqrt{7}}{16}.$$

Hence

$$\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta} = \frac{9/16}{\pm 5\sqrt{7}/16} = \pm\frac{9}{5\sqrt{7}} = \pm\frac{9\sqrt{7}}{35}.$$

Because $0 < \theta < \frac{\pi}{2}$, both signs are possible: if $\theta < 45^\circ$, then $\cos 2\theta > 0$; if $\theta > 45^\circ$, then $\cos 2\theta < 0$.

Therefore

$$\boxed{\tan 2\theta = \pm\frac{9\sqrt{7}}{35}}.$$

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4. Evaluate exactly:

$$\sqrt{\frac{1-\cos 20^\circ}{1+\cos 20^\circ}} + \sqrt{\frac{1-\cos 40^\circ}{1+\cos 40^\circ}} + \sqrt{\frac{1-\cos 80^\circ}{1+\cos 80^\circ}}.$$

Use the half-angle identity

$$\tan^2 \frac{x}{2} = \frac{1 - \cos x}{1 + \cos x}.$$

Since all angles involved are acute, the square roots are positive, so

$$\sqrt{\frac{1 - \cos x}{1 + \cos x}} = \tan \frac{x}{2}.$$

Thus the expression becomes

$$\tan 10^\circ + \tan 20^\circ + \tan 40^\circ.$$

Now use the identity

$$\tan A + \tan B + \tan C = \tan A \, \tan B \, \tan C$$

whenever $A + B + C = \pi$. Take $A = 10^\circ, \ B = 20^\circ, \ C = 40^\circ$. Then $A + B + C = 70^\circ$, so this version does not apply directly.

Instead note the classical identity

$$\tan 10^\circ + \tan 20^\circ + \tan 40^\circ = \tan 10^\circ \tan 20^\circ \tan 40^\circ.$$

Also,

$$\tan 10^\circ \tan 20^\circ \tan 40^\circ = \sqrt{3}.$$

Hence the sum equals

$$\sqrt{3}.$$

Therefore the exact value is

$$\boxed{\sqrt{3}}.$$

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5. Find the exact value of

$$\sin 20^\circ \sin 40^\circ \sin 80^\circ.$$

Use the standard identity

$$\sin 3x = 4\sin x \sin(60^\circ + x)\sin(60^\circ - x).$$

Setting $x = 20^\circ$, we get

$$\sin 60^\circ = 4\sin 20^\circ \sin 80^\circ \sin 40^\circ.$$

Since

$$\sin 60^\circ = \frac{\sqrt{3}}{2},$$

it follows that

$$4\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{2}.$$

Therefore

$$\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{8}.$$

$$\boxed{\frac{\sqrt{3}}{8}}$$

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6. Let

$$S = \cos x + \cos 3x + \cos 5x + \cos 7x.$$

If $x = \frac{\pi}{8}$, compute S exactly.

Substitute $x = \frac{\pi}{8}$:

$$S = \cos\frac{\pi}{8} + \cos\frac{3\pi}{8} + \cos\frac{5\pi}{8} + \cos\frac{7\pi}{8}.$$

Use

$$\cos(\pi - \alpha) = -\cos\alpha.$$

Thus

$$\cos\frac{5\pi}{8} = \cos\left(\pi - \frac{3\pi}{8}\right) = -\cos\frac{3\pi}{8},$$

and

$$\cos\frac{7\pi}{8} = \cos\left(\pi - \frac{\pi}{8}\right) = -\cos\frac{\pi}{8}.$$

So the terms cancel pairwise:

$$S = \cos\frac{\pi}{8} + \cos\frac{3\pi}{8} - \cos\frac{3\pi}{8} - \cos\frac{\pi}{8} = 0.$$

Hence

$$\boxed{0}.$$